Intelligent and Communicating Systems, ICS

2^{nd} Year Specialty SIQ G02

Lab n°05 -Annex-

Title:

IoT-Based Wireless LED Control Using eWeLink and Sonoff

Studied by:

First Name: Abderrahmane

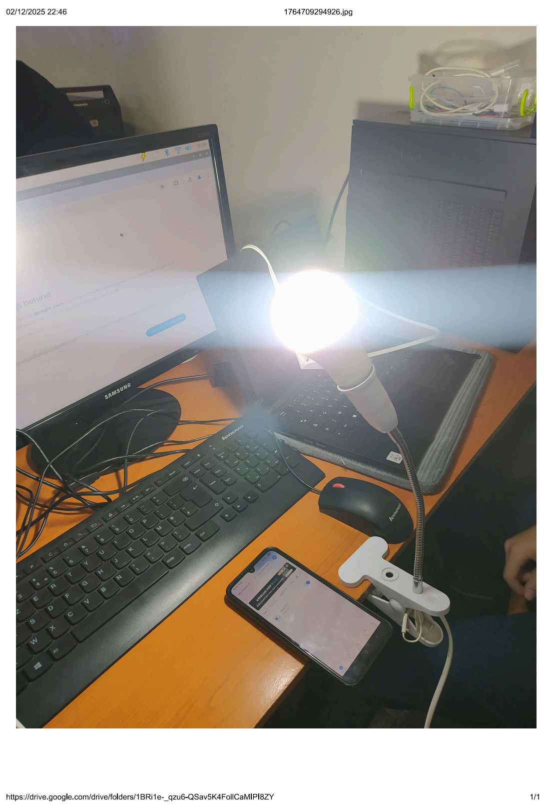
Last Name: GRINE

E_mail: ma_grine@esi.dz

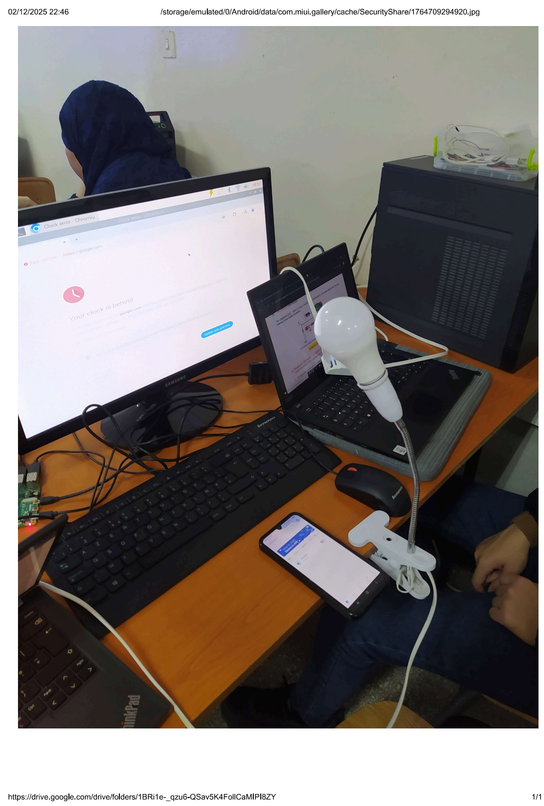
Annex: Wireless Control of an LED via eWeLink

A. Overview

This annex explains remotely controlling a pre-wired LED using the **eWeLink** app and a Sonoff Wi-Fi smart switch, demonstrating IoT integration in embedded systems.



(a) On mode



(b) Off mode

Figure 1: Led controll via eWeLink app

B. Required Hardware

- Sonoff Wi-Fi Smart Switch (e.g., *Basic R2*)
- 2.4 GHz Wi-Fi network
- Smartphone (Android/iOS) with eWeLink app

Figure 2: Sonoff Smart Switch

C. Wireless Setup

1. Installing eWeLink

- Install eWeLink from Google Play or App Store.
- Create an account using email or phone number.

2. Pairing the Sonoff Device

- Power on the Sonoff switch.
- Press and hold `SET` for 5 s until the LED blinks rapidly (quick pairing mode).
- In the app, select `Add Device` → `Quick Pairing`.
- Connect to a 2.4 GHz Wi-Fi network and assign a name (e.g., *LED Control*).

D. Remote LED Control

Once paired, the app allows:

- Toggle LED ON/OFF wirelessly
- Check real-time device status
- Schedule automatic ON/OFF events
- Use LAN mode for local control without internet (if supported)

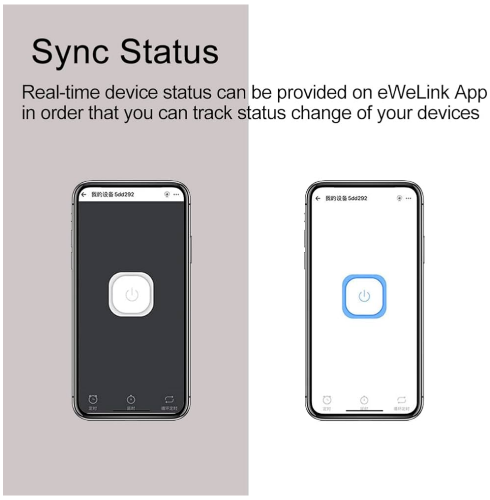


Figure 3: ON/OFF eWeLink user interface

E. Technical Notes

- Communication via eWeLink cloud using MQTT/HTTPS
- Relay electrically isolated from Wi-Fi module
- Latency typically <500 ms
- Safety features: overheat protection, authentication, encryption

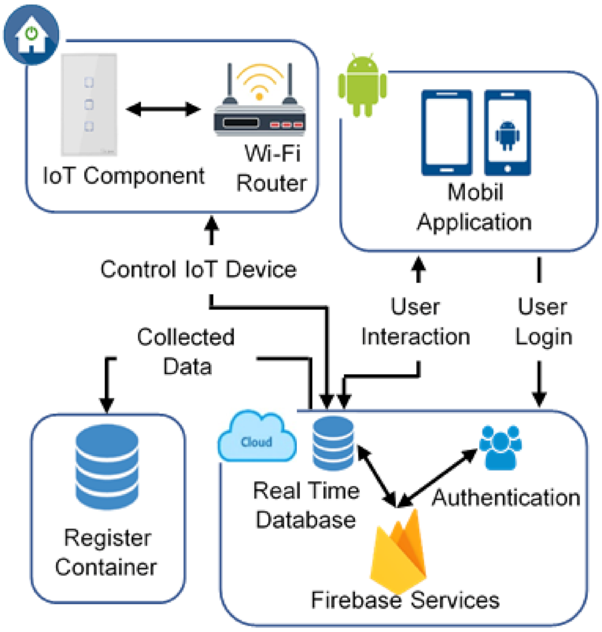


Figure 4: IOT smart home connection min-architecture

F. Conclusion

This setup demonstrates simple wireless control of an already connected LED using eWeLink and Sonoff, enabling IoT-based automation and cloud integration. This will serves as a simple initialization to the project of this module, where the Tuya application will be used.